M. Ahmad Siddiqui

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PROFESSIONAL SUMMARY

ML engineer (BS Data Science) who builds and evaluates LLM, RAG, speech and computer-vision systems that run on local hardware. Experience fine-tuning models with LoRA and QLoRA, building retrieval pipelines that verify their own answers, and deploying on-device inference with ONNX Runtime and llama.cpp. Reports results with held-out test sets, confidence intervals and documented failures.

EXPERIENCE

Rezinx *Remote*

ML Engineer (contract) *Aug 2025 – Present*

- Fine-tuned language models with LoRA for downstream production use cases.
- Built a recommendation engine with Superlinked and evaluated several embedding models for a job-classification task.
- Built a computer-vision hazard detector for fire, knives and stairs that estimates how close each one is to a child.
- Built a Grafana dashboard for monitoring models and systems.

Arch Technologies *Islamabad, Pakistan (Remote)*

Machine Learning Intern *May 2025 – Jun 2025*

- Built machine learning models to find patterns in large datasets, with data preprocessing and feature engineering to improve accuracy.
- Ran experiments to evaluate model effectiveness and recommended adjustments.

Ittehad Chemicals *Lahore, Punjab, Pakistan*

Intern *Jun 2026 – Jul 2026*

- Supported production and inventory processes while following plant safety protocols.

PROJECTS

PictureBOOK: offline picture-book generator *Rust, Tauri, SvelteKit, FastAPI*

- Built a desktop app that turns a typed or spoken story into an illustrated PDF in 6 to 9 minutes, running Whisper, Gemma 4 E4B and FLUX.2 klein in turn on one 8 GB GPU. Each model stage is an isolated, memory-capped process (tested: no stray process after SIGKILL).
- Benchmarked every stage (Whisper 1.9% WER on 48 synthetic clips; vision critic 94.7% accuracy on 40 synthetic images) and documented failures: no model detected extra limbs and 11 of 13 flagged defects survived redraws, so a human reviews every page. 156 Python tests.

Template_RAG: RAG library for documents and source code *Python, SQLite FTS5, Streamlit*

- Built hybrid retrieval (dense + BM25 + reciprocal rank fusion + listwise reranker), AST-based code chunking, and answer gates that require every quote to be found verbatim in the chunk it cites. 513 tests and CI.
- Evaluated on frozen gold sets with bootstrap confidence intervals and paired tests: the reranker added +18.6 recall@1 on 70 hand-written queries, while a web-trained cross-encoder never helped.

DLRAG: pre-registered comparison of RAG and tool-using agents *Python, llama.cpp*

- Designed 56 execution-graded coding tasks (24 frozen for the final test, sha256 manifest) and compared six retrieval and agent setups on a 2B model with paired bootstrap confidence intervals.
- Found that retrieval raised the full-pass rate from 3.3% to 33.3%, while tool-using agents added no detectable gain (-1.7 points, 95% CI -10.8 to +6.7) at about 3x the tokens.

QuestionPrep: AI interview practice platform (final-year project) *FastAPI, ChromaDB, llama.cpp*

- Built a local platform that generates interview questions from PDFs with grounding checks and scores answers in three stages (fuzzy match, BGE bi-encoder, DeBERTa NLI), with adaptive difficulty, token auth and SSE. 460 tests and CI.
- Implemented in-browser stress detection: a TCN on ONNX Runtime Web with a hand-written FFT, mel-filterbank and DCT-II front end, with no server round-trip.

Other projects

- Hybrid profanity detection and rewriting: LoRA-tuned DistilBERT (95.67% accuracy) routing to a QLoRA-tuned LFM2.5-1.2B rewriter via adapter switching. Edge-IIoT attack detection: multi-class intrusion classifier, 94.67% accuracy. PocketLaw: RAG chatbot over Pakistani banking law. Currency classifier: EfficientNet-B0 with text-to-speech for visually impaired users.

EDUCATION

Khwaja Fareed University of Engineering and Information Technology *Aug 2022 – May 2026*

Bachelor of Science in Data Science | CGPA: 3.35/4.00

SKILLS

ML & Deep Learning: Python, SQL, PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, LoRA, QLoRA, LLM fine-tuning, ONNX Runtime, llama.cpp

NLP & Retrieval: RAG, hybrid search (BM25 + dense), reranking, embeddings, ChromaDB, NLI, Docling, evaluation (recall@k, MRR, bootstrap confidence intervals)

Vision & Speech: YOLOv8, EfficientNet, transfer learning, object detection, OpenCV, Whisper

Backend & Tools: FastAPI, REST APIs, SQLite, Docker, Grafana, Superlinked, Git, pytest, GitHub Actions, Linux, Bash, uv